

A Study of Double-spending Causes and Countermeasures in Bitcoin Network

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Abstract—Bitcoin is a popular cryptocurrency based on a decentralized, tamper-resistant, and append-only ledger called blockchain, which researchers and industry have widely adopted. The security of Bitcoin heavily depends on a proof-of-work incentives-based consensus protocol. However, this protocol and its structure have some vulnerabilities. A double-spending attack is the most prominent and common threat to the Bitcoin protocol. The goal of this survey is to understand the various causes of and countermeasures to prevent a double-spending attack. In this paper, we present a state-of-the-art double-spending attack survey on the Bitcoin network. We emphasize the gap associated with the blockchain protocol, its delay-causing factors, and protocol features exploited to double-spend Bitcoins. We study both theoretical and experimental work published in recent years. Finally, we discuss existing countermeasures and a few open issues in identifying and tackling a double-spending attack in the Bitcoin network. Our results can help Bitcoin users determine a trade-off between the risk of double-spending attempts and transaction confirmation delay.

Index Terms—Bitcoin protocol; Double-spending attack; PoW consensus; Blockchain.

1. Introduction

Blockchain is a cryptographically secure payment system that works without a trusted third party in a peer-to-peer network. Bitcoin is the first widespread cryptocurrency deployed on the blockchain. Over time, researchers added many safety measures or security aspects to the basic Bitcoin protocol. Security plays an important role in guaranteeing blockchain acceptability. The Bitcoin protocol has many vulnerabilities that researchers have not conquered completely. Double-spending is the most prominent and common threat which exploited these vulnerabilities in bitcoin [1] [2].

Various delay-causing factors such as transaction or block propagation, proof-of-work consensus, and transaction or block validation process, etc., lead to inconsistent views of the ledgers in the bitcoin network [3], [4]. One of these factors could be exploited to affect the consensus protocol adversely and, if used correctly by an attacker, could lead to a successful double-spending attack.

The current Bitcoin protocol has solved the double-spending problem by waiting for many block confirmations [5]. It is well suited for slow payment transactions. As the number of confirmations increases, the chances of a successful double-spending attack decrease. There are different ways to double-spend bitcoins using unconfirmed or confirmed transactions discussed in Section 2. Recently, there has been an increase in research efforts from various groups providing new strategies against the double-spending problem in micro-payment transactions. A few survey papers discuss prevention strategies with certain assumptions for double-spending attacks under different scenarios [6], [7], [8], but they have not covered the vulnerabilities of protocol features. Some of the prior surveys covered attacks and vulnerabilities, but they do not provide the details of exploited protocol features [7], [9], [10], [11]. We focus on double-spending related attacks, exploited protocol features, associated vulnerabilities, involved entities, and existing countermeasures. Our primary focus is to understand the causes of double-spending-related attacks in great detail to help improve the Bitcoin protocol. We have compared the existing literature with the current work in Table 2.

1.1. Our contribution

- We identified vulnerabilities that attackers could exploit to double-spend bitcoins.
- We provide a systematic overview of double-spending related attacks launched due to vulnerabilities in the protocol features.
- We classify these attacks, vulnerabilities exploiting protocol features, and entities involved in launching these attacks.
- We compare existing countermeasures and their underlying defensive strategies.

1.2. Paper Outline

The rest of the paper is organized as follows. Section 2 provides an overview of the Bitcoin blockchain. More specifically, transaction lifecycles, major protocol modules, blockchain forks, and double-spending risks in slow/fast

payment transactions in the Bitcoin blockchain are discussed. We present the research strategy and related work in Section 3. The analysis of double-spending risk, including various related attacks and their classification listing vulnerabilities, exploited protocol features, and involved entities, are covered in Section 4. We discuss the existing countermeasures and strategies for double-spending related attacks in Section 5. The subsequent Section 6 summarizes critical observations from our survey and future directions. Finally, we conclude the paper in Section 7.

2. Overview of Bitcoin Blockchain

This section introduces the Bitcoin protocol and double-spending risk. We first present the propagation-synchronization process between the nodes and then the proof-of-work-based consensus mechanism. Later, we briefly talk about double-spending risk and provide the related work.

2.1. Bitcoin Protocol

When a new node boots up in the Bitcoin network, it must discover at least one existing node. A node must connect to it to participate. Various methods are used to find the peers. One of them is to query DNS seeds or servers that provide a list of IP addresses of Bitcoin nodes. On the other hand, a bootstrapping node that knows nothing of the network must be given the IP address of at least one Bitcoin node, and it establishes a connection for further communication. The nodes establish a TCP connection to connect to a known peer, mostly on port 8333 [12]. A node starts a handshake upon establishing a connection by transmitting a version message to check if the remote peer is compatible. Later, they send address and getaddress messages to receive the peers' IP addresses and their peers' addresses as well. Upon losing old connections, a node must continue discovering new nodes and assisting other nodes when they bootstrap. Once bootstrapping is over, a node remembers its most recent successful peer connections. This is required if it is rebooted; then, it can quickly re-establish connections with its former peers. The node uses seed nodes to bootstrap again in case none of the former peers responds to its connection request [13].

When a node having wallet software "receives bitcoins," it means that the wallet detects unspent transaction outputs (UTXOs), and such UTXOs can be spent with one of the keys controlled by the wallet. Thus, a user's balance is the sum of all UTXOs that the user's wallet can spend. The wallet scans the blockchain and aggregates the value of any UTXO to calculate the user's balance. Most wallets maintain a database or use a database service to store a quick reference set of all the UTXO they can spend with the keys they control. The wallet software collects UTXO, provides the appropriate unlocking scripts, and constructs new outputs assigned to a recipient to create a transaction. This resulting transaction is then sent to the peer nodes using a gossip communication protocol so that it can be

propagated across the entire Bitcoin network. Every Bitcoin node that receives a transaction verifies the transaction and then forwards it to its neighbors. It ensures that only valid but unconfirmed transactions are stored in the local memory pool and are propagated across the network. In contrast, invalid transactions are discarded by the first node that encounters them.

Later, a mining node aggregates transactions from the local memory pool and creates a candidate block. Then, mining nodes try to find a nonce to solve the proof-of-work (PoW) puzzle for the newly created block [13]. The PoW mechanism consists of finding a hash of the block header containing the nonce and checking if it is smaller than the current difficulty target. The difficulty target is dynamically adjusted to maintain an average block generation interval of 10 minutes. There is a race between the mining nodes to solve this PoW puzzle. The node that finds a valid nonce first appends the newly mined block to its local copy of the blockchain and propagates it further into the Bitcoin network using the same gossip protocol. The block that is accepted by the majority of the Bitcoin network wins the race [14]. The block owner can claim a certain amount of tokens as a reward for mining the block; further, it can collect fees from all enclosed transactions.

The transaction is "confirmed" when k more blocks are appended to the block containing the transaction [15]. Every new block is either accepted or rejected by any honest node. There is a chance of discarding an accepted block, but with exponentially diminishing probability as the blockchain grows. The bitcoin transaction's lifecycle, from transaction creation, till it gets mined into some block, is shown in Figure 1. It has five stages: Transaction creation, propagation, verification, mining, and confirmation. We have identified two basic protocol modules, propagation & synchronization, and mining, which cause delays in transaction confirmation. An attacker can take advantage of this to perform a double-spending attack.

Let us understand the delay-causing factors associated with each module in detail.

Propagation and Synchronization:

The foundational infrastructure of blockchain is a peer-to-peer overlay network on top of the Internet. On start-up, when a node joins the Bitcoin network, it needs to synchronize with the current state of the blockchain. The time needed is referred to as "bootstrapping time," which mainly depends upon node connectivity. Every node (or peer) in the Bitcoin network needs to propagate information like transactions or blocks to its randomly selected peers. Each node can connect to at most eight outbound nodes (peers) and up to 117 inbound peers [16]. Outbound peers are the nodes that our node needs to find in the Bitcoin network, while inbound peers are the nodes that find our node through the network. Both of these connections use the TCP transport protocol. Some of the Bitcoin nodes have fewer connections, called lightly connected, whereas nodes with more connections are called highly connected nodes. High connectivity ensures fast propagation of transactions and blocks throughout the network. In general, any node

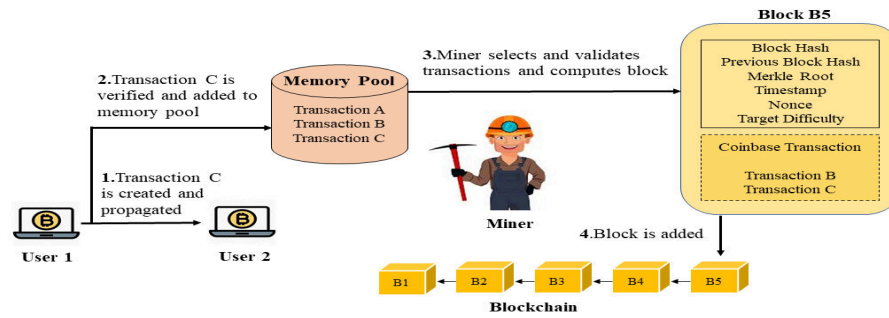


Figure 1: A Lifecycle of the Bitcoin Transaction

can have at most 125 total connections.

The transaction and block delivery protocol follows a two-way handshake over each TCP connection. The source node sends an *inv* (inventory) message about a transaction or block to its direct neighbors. The receiving node sends a *getdata* request if it does not have the transaction/block; otherwise, it ignores the message. Then, the actual payload is transferred by the source node. This method of propagation is referred to as advertisement-based propagation [17]. In “unsolicited push propagation,” nodes skip sending *inv* message and *sendheader* messages; instead, it sends the transaction/block to other nodes directly. This method improves the speed of propagation [18] [19]. Before propagating a transaction or block, the nodes need to validate them. The transaction verification process includes data format checking, digital signature verification, and whether the output amount is smaller or equal to the transaction’s sum of inputs. The node also checks whether transaction outputs have already been spent. On the other hand, block validation means checking whether the block header’s hash, along with the nonce, is smaller than the target difficulty or not, as well as verifying all transactions contained inside the block. The time required for validation depends on the node type, block size, and UTXO set search time.

Block Mining:

The node’s memory pool stores the transactions received from the neighbors. Mining nodes give priority to the transactions with high transaction fees and consider such transactions for mining. Sometimes, transactions to be mined are selected randomly from the memory pool. Due to the fixed block size, each arrived transaction has some non-zero waiting time [20]. Similar to transactions, nodes receive mined blocks from their neighbors. If a new block arrives while the node is engaged in solving the PoW, it can waste hashing power and time in solving the PoW. Therefore, the mining activity depends on the arrival rate of transactions and blocks. It also depends on the memory pool’s waiting time for block creation and hashing power of the mining nodes [21].

In this way, the factors listed above in both these modules delay the transaction confirmation process.

2.2. Blockchain Forks

The regular operation of the Bitcoin blockchain assumes that all nodes have an identical copy of the ledger. Every copy of the ledger has the same block in the head portion of the blockchain. Blocks get mined asynchronously across the mining nodes. Due to the network delays, it is unlikely that the same block will be found in the head portion of the ledgers maintained locally by multiple nodes.

Let us assume that node *A* has mined block *P* and starts forwarding it to the rest of the Bitcoin network. At the same time, another mining node *B* mined a block *Q* and forwarded it to its neighbors. Block *Q* may have a partial or complete set of transactions overlapping with *P*, except for a Coinbase transaction that allocates the reward to the mining node *B*. In this case, some nodes accept block *Q* as the head of their main chain and block *P* as the side chain’s head, while other nodes do the reverse. This inconsistency in local blockchain copies across the network creates a blockchain fork. Here *P* and *Q* have the same parent block, and a blockchain fork gets created. Likewise, multiple forks are possible in a real scenario. A node can’t figure out which block is legitimate and must retain both blocks until the fork resolves. However, competing blocks may contain contradicting transactions that use the same input and perform a double-spending attack [3], [4].

According to its previous block hash entry, the newly mined blocks, either local or received from neighbors, get appended to the corresponding chain. Subsequently, nodes adopt a chain that has the most cumulative “Chainwork,” where Chainwork refers to the total number of hashes computed in mining a block. Therefore, the chain that grows longer in terms of the chainwork and has taken the most energy to build a block wins the mining race.

Finally, there remains only one main chain that is consistent across all Bitcoin nodes. This is the traditional way

to avoid forks in the Bitcoin blockchain. The number of blocks appended over any block is called the number of block confirmations. Various consensus protocols like PoS, DPOS, and PBFT reduce the chances of blockchain forks [22], [23], [24], [25].

2.3. Double-Spending Risk

The most concerning security risk in the Bitcoin blockchain is the double-spending risk. In the above subsection, we have seen how a fork causes a double-spending attack. Let us understand double-spending attack scenarios involving slow and fast payment transactions that most existing literature has considered for evaluation. An attacker attempts to make two conflicting transactions using the same input UTXOs. The attacker sends the genuine transaction TX A to the network and starts secretly mining a block containing the fraudulent transaction TX B, as shown in Figure 2. Once the genuine transaction is confirmed, the

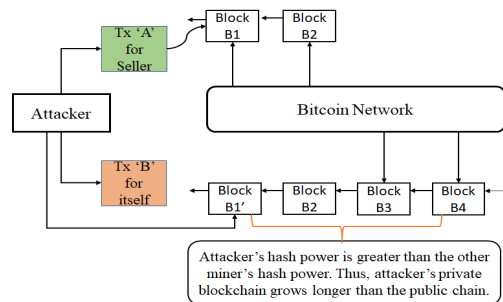


Figure 2: A diagram of a double-spending attack in slow-payment transactions. In this diagram, an attacker sends transaction A (TX A) to the seller and transaction B (TX B) using the same bitcoins to the address he owns. TX A gets mined in block B1, and TX B gets mined in block B1'. Later, the majority of the network accepts the chain containing TX B and becomes longer than the chain containing TX A.

attacker gets the service against it [26]. The attacker immediately releases his secretly mined blocks and ensures that a chain containing the fraudulent transaction is longer than the chain containing a genuine transaction. The bitcoin protocol accepts the longest chain [27]. Thus, it forces the Bitcoin network to accept a fraudulent chain and abort the genuine transaction chain. Here, the attacker's hash power is higher than that of the other miners in the network, leading to the blockchain fork. Thus, the double-spending attack becomes successful in slow-pay transactions where payment is made after a transaction is confirmed. On the other hand, in a fast-pay transaction, the seller gives service immediately after seeing the transaction in the network without waiting for its confirmation [28], [29]. In Figure 3 an attacker sends transaction A (TX A) to the seller and transaction B (TX B) using the same bitcoins as those involved in TX A's input, to colluding helpers. The attacker is required to trick the victim into accepting TX A that the victim will not be able to redeem subsequently. The seller does not receive TX B due to colluding helpers, and the seller provides service to an Attacker. On the other hand, colluding addresses help

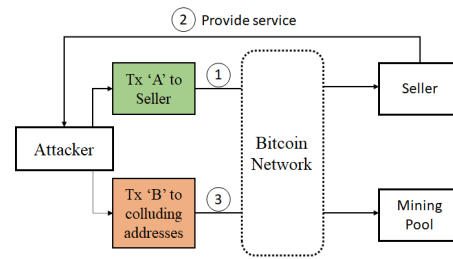


Figure 3: A diagram of a double-spending attack in fast-payment transactions. In this diagram, an attacker sends transaction A (TX A) to the seller (1) and receives service from the seller (2). Later, the attacker sends transaction B (TX B) to his colluding partners (3). These colluded partners send TX B to the mining pool such that TX B gets mined and confirmed on the blockchain.

mine TX B by propagating it to mining pools faster. These transactions are released at the same time or at different times into the bitcoin network based on the attackers' strategy. If both transactions are released simultaneously, they are likely to have equal chances of getting confirmed in an upcoming block. A bitcoin node accepts the transaction that reaches it first and ignores all subsequent transactions. If TX B gets confirmed on the blockchain, then the double-spending attack succeeds. Fast payments build upon trusting that a transaction will eventually be confirmed. Such fast-pay transactions are prone to double-spending attacks [30]. Attackers are continuously trying to find new methods to infiltrate into the Bitcoin network to perform a double-spending attack [31], [32].

Bitmex research reported on January 21, 2021 in the "Business Insider," financial news channel that small double-spend of around 0.0006206 BTC (21 USD) was detected but later Bitfinex CTO, Paolo Ardoio, said that double-spending event did not occur — the payer had replaced the transaction by giving more fees. It just happened that both transactions were mined in two separate blocks. Eventually, one of the blocks was accepted by the Bitcoin network [33].

3. Literature Search Strategy

In this article, we focus on protocol vulnerabilities and causes of double-spending attacks and countermeasures to prevent them. We followed a systematic literature review (SLR) process [34] [35] to collect papers to survey.

3.1. SLR Process

We have gone through studies related to double-spending attacks in Bitcoin. The causes for double-spending attacks, existing countermeasures to prevent them, the combined DS attacks and the defenses, open challenges, and future research directions are outlined in the paper. While performing SLR, the steps are: Formulation of key research questions, the search process, criteria for selecting articles and the data extraction process, and the execution of analysis and classification.

TABLE 1: Data Sources

Source IDs	Data Source Name	URL
D1	IEEE Xplore	http://ieeexplore.ieee.org/
D2	ACM	http://dl.acm.org/
D3	Science direct	http://sciencedirect.com
D4	Springer link	http://link.springer.com/
D5	Google scholar	http://scholar.google.com

A. Research Questions

The following research questions initiate the SLR:

- RQ1: Which attacks affect the Bitcoin network and cause double-spending?
- RQ2: Which of the protocol features are being exploited for double-spending bitcoins? In other words, what are the loopholes in the existing protocol leading to double-spending?
- RQ3: Which solutions are mainly used to counter double-spending-related attacks? Do existing countermeasures suffice?
- RQ4: How can one categorize existing solutions to DS attacks?

B. Keywords

We considered the following keywords, as we focused only on double-spending related attacks in Bitcoin. To extract the relevant attacks, we used the following terms: “double-spending attack bitcoin,” “bitcoin security,” “causes of and countermeasures to prevent double-spending attacks,” “detection and prevention double-spending,” “double-spending related attacks,” and “data inconsistency attack in bitcoin.”

C. Research Filtration Process

Here, we outline the process to extract relevant information for this survey. Table 1 indicates the source of literature and its URL:

Search strategy: The process starts by collecting papers based on the keywords mentioned above. We followed the review strategy, as shown in Figure 4. We utilized some keywords from our selected data sources for primary studies. The first step in the filtration process is to filter the literature by reading titles and abstracts. Exclude papers that are unrelated to double-spending issues or not published in English. The second stage of filtration is to filter any article that is not related to Bitcoin double-spending attacks. The third filtration stage is to verify the filtered articles and check whether they have been peer-reviewed or not. The final stage is to read the papers in full and exclude any that are not explicitly related to this research domain, double-spending related attacks in Bitcoin.

3.2. Related Work

The acceptance of blockchain technology continues to grow in various applications like IoT, healthcare, medical, finance, and smart vehicles. This growth of blockchain-based solutions maximizes the importance of blockchain security. A few research studies evaluated bitcoin security

– mainly double-spending – but they fall short in various aspects. We have addressed these shortcomings and presented them in Table 2. Nicolas et al. [6] performed a

TABLE 2: A Summary of related work that explores the security threats leading to double-spending attack. (✓) denotes that the stated characteristic was addressed, while (-) denotes that it was not addressed.

Reference Surveys	DS Attack	Vulnerabilities	Entities Involved	Countermeasures	Prevention Strategies
Nicolas et al. [6]	✓	-	-	✓	✓
Dasgupta et al. [9]	✓	✓	-	-	-
Saad et al. [8]	✓	-	-	✓	-
Zhang et al. [36]	✓	-	-	✓	-
Pinzon et al. [37]	✓	-	-	-	-
Mosakheil et al. [10]	✓	✓	-	-	-
Conti et al. [7]	✓	✓	✓	✓	-
Li et al. [11]	✓	✓	-	-	-
Current Work	✓	✓	✓	✓	✓

literature study to explore countermeasures against double-spending and selfish mining attacks. Their study identified six defensive strategies, which we have also referred to in Table 5. Conversely, we considered almost all variants of double-spending attacks, associated vulnerabilities in protocol features, and countermeasures to reduce the risks. The authors in [9] studied threats and related vulnerabilities in great detail, but they have not covered the exploited protocol features and countermeasure strategies. The authors in [8] identify different attacks on blockchain cryptographic constructs, peer-to-peer networks, and blockchain-based applications. Their study mainly focused on attack surfaces, but they did not discuss the threats and their associated vulnerabilities in detail. Zhang et al. [36] present the attack model that combines a double-spending attack with a Sybil attack in the Bitcoin network. Their study discussed countermeasures but did not provide information about root causes or vulnerabilities associated with the protocol similar to [8]. Pinzon et al. presented two different double-spending attack models with time advantage and gave the probability of a successful double-spending attack. The research work has not presented the attack scenarios, vulnerabilities, and countermeasures [37].

Another work by Mosakheil et al. [10] has not explored the threats in detail. Their work does not provide detailed information about attacks and vulnerabilities associated with the protocol. Conti et al. [7] referred to various threats in detail, along with the existing countermeasures. However, the authors have not discussed the defensive strategies and exploited protocol features. On the other hand, [11] discusses the attacks and associated vulnerabilities, but they have not covered the exploited protocol features and defensive measures against security threats. These related studies fail to explain the threats, the vulnerabilities exploited, associated protocol features, countermeasures, and strategies in one place. No study has discussed the combined double-spending related attacks. In this work, we studied double-spending related attacks, the vulnerabilities, the protocol fea-

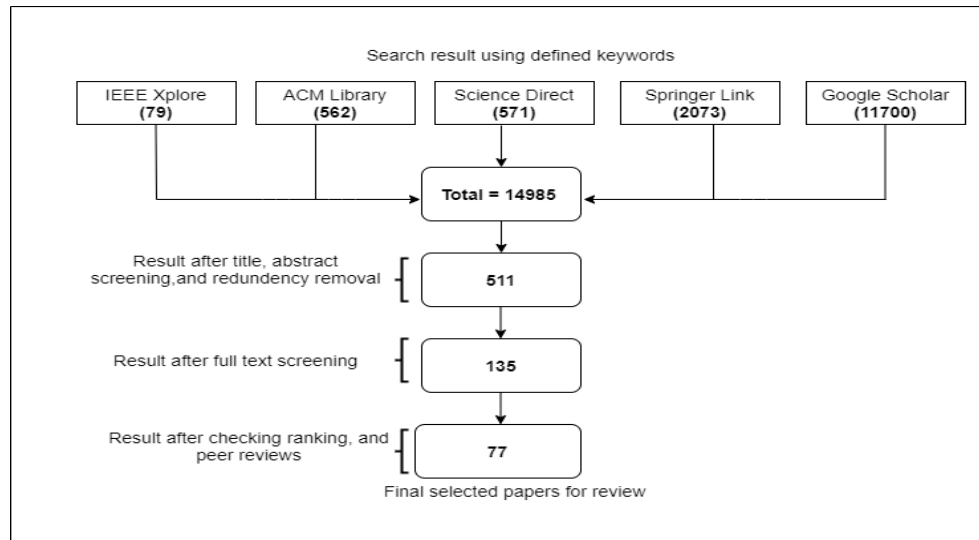


Figure 4: Review Strategy

tures to be secured, combined attacks, and countermeasures for reducing the risk in the bitcoin blockchain.

4. Analysis of Double-Spending Risks

This section explains how to perform a double-spending attack, followed by a description of all related attacks. Later, we provide a classification of double-spending attacks. A double-spending attack can be performed by exploiting the vulnerabilities associated with Bitcoin protocol modules. The main modules are propagation-synchronization and mining-based consensus. After studying all related attacks, we have observed only two ways to perform a double-spending attack. The first method is to delay regular transactions so that the majority of the network accepts fraudulent transactions. Another way is to secretly mine a block containing a fraudulent transaction and release a block such that it forms a fork and wins the race. In both these methods, an attacker would like to get the service from the seller; until then, he prevents his fraudulent transaction or block containing fraudulent transactions from reaching the seller.

We have shown this classification in Figure 5.

4.1. Attack cases

Double-spending related attacks help an attacker gain monetary benefits in the blockchain. Here, we discuss in detail all such attacks that lead to double-spending bitcoins.

4.1.1. Attacks Exploiting Propagation and Synchronization Module.

- Race Attack:

The Race attack spends the same bitcoins in multiple transactions. It is caused by sending two conflicting transactions in rapid succession. The malicious user sends a genuine transaction to the Seller. On the other

hand, he sends a fraudulent transaction to the rest of the Bitcoin network. The Seller node gives the service for an unconfirmed transaction to the malicious user; however, the fraudulent transaction gets confirmed on the blockchain. Therefore, the user must wait for some minimum number of confirmations before accepting a transaction as valid [5], [26], [38], [39], [40], [41].

- Sybil Attack:

In this attack, the malicious user could set up multiple instances of him or dummy helper nodes to confirm the transaction and compromise the Bitcoin network. In the Sybil attack, a group of fake nodes gives the majority of confirmations to the fraudulent transaction. But proof-of-work prevents Sybil attack, because verification depends on computational power and not on the fake identities [42], [43].

- Balance Attack:

This attack consists of a subgroup of miners with balanced hash power. The malicious user can delay the network communication between these subgroups of miners, which leads to the double-spending attack [44].

- Eclipse or Net split Attack:

The malicious user can prevent regular miner activity by blocking the miner's incoming and outgoing connections or diverting incoming transactions to himself. This separates a miner from the rest of the Bitcoin network – this activity is called Network Partitioning. With this approach, a malicious user can perform a k -confirmation Race attack [17], [45], [46].

- Tampering Attack:

Tampering Attack helps to delay the propagation of transactions or blocks to the specific node. Helper nodes colluding with the malicious user delay propagating a genuine transaction meant for other nodes in the Bitcoin network [42].

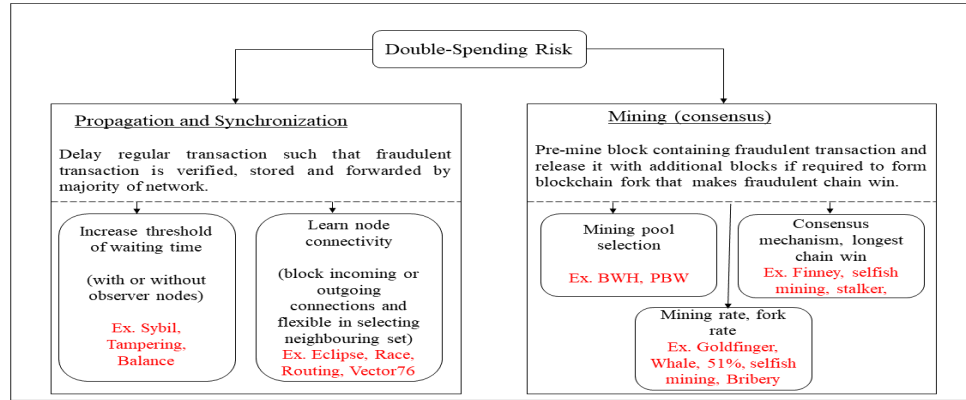


Figure 5: The Classification of Double-spending Risk

- **Routing Attack:**
A routing attack separates a subset of nodes from the Bitcoin network that monitors the round-trip time for block propagation and increases diversity in node connections; this helps in delaying block propagation. Such activities help the malicious user to perform a 0-confirmation Race attack [47], [48], [49].
- **Vector76 Attack:**
This attack is a combination of a Race attack and a Finney attack where the Seller provides service to the malicious user after accepting 1-confirmation of transaction. This attack is also known as 1-confirmation Attack [50].

4.1.2. Attacks Exploiting Mining Module.

- **Finney Attack:**
The malicious miner broadcasts a pre-mined block as soon as it receives the Seller's product for launching a double-spending attack [51].
- **Brute Force Attack:**
A Brute Force attack is an extension to the Finney attack where the Seller waits for z number of confirmations. The malicious user privately does mining on blockchain fork to perform a double-spending attack [52], [53].
- **Goldfinger Attack or $> 50\%$ Attack:**
A single miner or mining pool controls more than 50% of computing power in the network, which is the worst possible scenario in the Bitcoin network. A malicious user controls the subset of miners who form a mining pool or collude with a miner who has $> 50\%$ of computational resources to double spend the bitcoins [54].
- **Selfish Mining Attack:**
Malicious miners hide the mined blocks' details and reveal them if they obtain an unfair reward greater than spent computing power [1]. A malicious miner confuses other miners and makes them waste their resources on incorrect blocks. Later, the selfish miner

releases the secretly mined block to disseminate a fraudulent chain that is longer than the honest chain. This is similar to a Race attack with k -confirmations [55].

- **Bribery Attack:**
The malicious user bribes miners to mine a block on his behalf. The mined block contains a fraudulent transaction [56]. Minority miners can double-spend bitcoins by giving incentives to the group of rational miners. Such an attack is known as a Whale attack [57].
- **Block Withholding Attack:**
Here, the attack is performed on a mining pool where a pool member never publishes full proof-of-work; instead, he gives partial proof-of-work. The malicious user node does not gain any bitcoins, but it just makes other pool members lose, or it performs a complex block-concealing attack like selfish mining [58], [59], [60].

4.2. Classification of Double-spending Related Attacks

We have presented all double-spending related attacks, vulnerabilities, and associated protocol features in Table 3 and Table 4. There are various factors associated with the bitcoin protocol, such as gossip communication, transaction or block validation process, bootstrapping process, etc. These factors are the vulnerabilities associated with the protocol that are exploited to launch an attack. We have also added a column listing the entities involved in these attacks, like blockchain, users, applications, miners, and mining pools.

4.3. Discussion

The default solution [26] that deters double-spending in bitcoin is its use of the Proof-of-Work (PoW) based

TABLE 3: Classification of Attacks Exploiting Vulnerabilities in Propagation & Synchronization

Propagation and Synchronization Module				
Attacks	Description	Protocol Feature Exploited	Entities Involved	References
Race (k -confirm)	exploit intermediate time between transaction initiation and confirmation	gossip communication protocol, transaction or block validation time, node types	Blockchain, Users	[5], [26], [37], [38], [39], [47], [66], [67], [68], [69]
Sybil	delay block propagation, increase the threshold of waiting time	gossip communication protocol, node identities, transaction/block validation time	Users	[42], [43]
Balance	delay network communication between subgroup of miners	gossip communication protocol	Application, Users	[44]
Eclipse or Netsplit	learn node connectivity and flexibility in selecting neighbor set, direct connection to incoming packet	bootstrapping process, node connectivity	Miners, Users	[45]
Tampering	learns node connection or network topology	bootstrapping process, gossip communication protocol	Blockchain, Users	[42]
Routing	learns node connection or network topology	gossip communication protocol, transaction/block propagation, node connectivity, routing time	Application, Users	[47], [48]
Vector 76	maintain static IP address list and accepts direct incoming connections, accepting 1-confirmation transaction	gossip communication protocol, transaction/block propagation	Application, Users	[50]

This table details attacks related to the propagation-synchronization module, actions performed by an attacker to launch the attack, exploited protocol features, entities involved, and reference papers. The literature listed in each row has studied various attacks exploiting vulnerabilities related to one or more factors associated with the protocol.

consensus algorithm, which imposes a significant computational burden on a malicious user who wishes to double-spend. In [61], the analysis failed to consider the varying hash rate and block withholding attack, which Rosenfeld analyzed in [5]. Rosenfeld could not provide the expression for the number of confirmations required to reduce double-spending attack chances to negligible levels. Sompolonsky and Zohar in [55], [62], [63], [64] proved that Nakamoto's claim regarding propagation delay (i.e., propagation delay is negligible in the bitcoin network) is not correct. The transaction processing rate is a function of bandwidth, and network delay causes the propagation delay [65]. Later, the authors proposed GHOST (Greedy Heaviest-Observed Subtree), a new policy for chain selection that increased the transaction processing rate. They considered Finney, Vector 76, and selfish mining attacks in their work. For the first time, a lower bound on waiting time for giving service is provided by Gervais et al., who detected and prevented the DS attack [42].

In 2015, Mathias and Decker witnessed the race attack by exploiting knowledge of the network topology [68]. According to them, peer identification helps form a successful double-spending attack with high probability. The chances

of a successful double-spending attack increase from 12 to 60% if the topology is known. Pinzon and Rocha in [37] analyzed and detected the race attack and block withholding attack by using the same strategy of varying hash rate and "longest chain wins" mechanism as Rosenfeld. However, they failed to give proof of the exponential decay function in finding the probability of a double-spending attack. For the first time in [39], the authors presented a closed-form expression and the proof for a successful race attack, but it lacks the formal testing of the solution in the Bitcoin P2P network.

In 2016, [67] proposed dynamic policies for transaction acceptance that depend on both the amount of time since the transaction broadcast and the number of confirmations. However, their analysis was not for any general transaction over any time window. Later, there was a proposal on reducing propagation delay where the authors invented a clustering-based ping time protocol that creates node clusters based on locality and node connectivity [47]. They identified a distance threshold based on which transaction propagation delay is improved. One needs to measure the ping latencies between the nodes and group nodes into clusters. There was an attractive solution by Alexandra

TABLE 4: Classification of Attacks Exploiting Vulnerabilities in Mining

Mining based Consensus Module				
Attacks	Description	Exploited Feature of the Protocol	Entities Involved	References
Finney	accepts 1-confirmation transaction	mining rate, block confirmation	Miners, Mining pool, Users	[51], [62], [63], [64]
Bruteforce	flexibility in number of block confirmations	mining rate, block confirmations	Miners	[52], [53], [70]
Goldfinger or 51%	holds large portion of computing power, inappropriate consensus mechanism	p2p network and mining	Blockchain, Miners	[54], [71]
Selfish Mining	increases waiting time of regular transaction confirmation	mining rate, longest chain mechanism in case of forking	Blockchain, Miners, Mining pool	[4], [55], [72]
	stalker attack - variant of the selfish mining which prevents a given node from publishing a block based on height of fork	fork rate, longest chain mechanism	Blockchain, Miners, Mining pool	[73]
Bribery, Whale	provides incentives flexibility to rational miners	mining rate	Miners	[56] [57]
Block-Withholding	delays or drop valid blocks	p2p network and mining	Miners, Mining Pool	[58]
	PBW attack - variant of the block withholding attack where attack is launched by pool manager	mining pool selection	Miners, Mining pool	[74] [75]

This table details attacks related to the mining module, actions performed by an attacker to launch the attack, exploited protocol features, entities involved in this attack, and reference paper. The literature listed in each row has studied various attacks exploiting vulnerabilities related to one or more factors associated with the protocol.

et al. in 2017, who said that a double-spending attack is possible under the offline setting. They proposed an offline wallet. But these offline wallets certify themselves with a Certification Authority; this deviates from Blockchain's decentralized nature [69]. Goffard analyzed the DS attack and selfish mining attack and provided an upper bound on the number of confirmations [72]. In [76], the authors analyzed the attack from [38] and provided a closed-form expression for a successful race attack.

We also studied researchers' theoretical work on finding the required number of confirmation blocks before giving service to the Buyer. The original analysis in [26] assumed that an attacker's number of blocks follows a Poisson distribution. In contrast, [5] considered a negative binomial distribution. Later, in 2016, the same authors found a new dynamic policy. They have shown that the probability of the double-spending attack is a function of the number of confirmations and the time elapsed since the transactions were broadcast. They have analyzed the Race Attack, Finney Attack, and 50% Attack using the formalism of Markov Decision Processes. Blocks found by an attacker follow a random walk with a limiting distribution [67].

The hash rate-based model for analyzing a pre-mining attack is not enough, as Pinzon and Rocha claimed in 2016 [37]. They proposed a time-based attack model for the Race attack and the Block Withholding Attack, where the time required for finding the blocks by an attacker

follows the Erlang distribution. For the first time, [39] gave an asymptotic formula for exponential decay and closed-form expression of the race attack probability. The authors analyzed race attack and selfish mining and gave a closed-form formula for the likelihood of double-spending in terms of the Regularized Incomplete Beta (RIB) function. Later, in early 2018, authors in [77] extended the work done by Sompolonsky and Zohar. They gave a closed-form expression for the double-spending attack probability using a finite sum and claimed that dynamic policy reduces the average waiting time for transaction acceptance. They have studied the security and performance properties of Dynamic Policy, which was missing in Sompolonsky's paper, and used a continuous-time Markov chain to analyze the race attack.

Authors in [76] assumed a negative binomial distribution for the number of blocks generated by the attackers. The authors gave a closed-form expression for the probability of double-spending, the expected reward for a malicious miner. They claimed that their adaptive double-spending attack was much stronger than the double-spending attack proposed by Nakamoto. Further, in 2019, Ramezen et al. analyzed an adaptive double-spending attack using a continuous Markov process [76]. In 2021, authors in [78] presented an adaptive DSA attack. The authors used a Markov decision process in their analytical model. They have also utilized stochastic dynamic programming to obtain optimal attack strategies. There were only two solutions after Nakamoto and Rosen-

feld analyzed the bounds on the number of confirmations sellers must wait for before providing service to the buyer, as proposed by Sompolsky '16 and Goffard '18.

5. Existing Countermeasures

This study includes collecting theoretical and experimental results for securing the blockchain from a double-spending attack. It will help readers to identify new defensive strategies against double-spending-related attacks. We have referred to the work by Nicolas et al. [6] for countermeasure strategies as shown in Table 5.

TABLE 5: Double-spending prevention strategies [6]

Strategy	Description
Monitoring	Countermeasures that monitor users activity depending upon time, usage, and tasks.
Forwarding	An alert system that aims to notify attacks to the network nodes.
Broadcasting	Provide public system information to recipient addresses, for example, private details relating to attackers.
Detection	Inspection that detects and notifies about misbehavior in the bitcoin blockchain.
Conceptual design	Relates to conceptual research proposal and theoretical proposed techniques.
Inform	Techniques to analyze bitcoin security based on raw facts, models, figures, processes, and mathematical algorithms.

Further, we summarised the existing countermeasures against double-spending related attacks discussed in Section 5 and their countermeasure strategies. The results from our study are shown in Table 6 and 7.

5.1. Double-spending Countermeasures Review

Bitcoin prevents the Race attack by enforcing a simple rule that only unspent outputs from the previous transaction can be used in the next transaction's input. Their chronological order specifies the order of transactions in the blockchain, enforced using strong cryptographic techniques. It boils down to a distributed consensus algorithm and time-stamping. Nakamoto [26] and Rosenfeld et al. [5] suggest increasing the number of confirmations to decrease the chances of double-spending attacks. However, Karame's proposal suggested inserting observers in the network and communicating alerts to help sellers detect double-spending attacks [31], [32], [52]. Enhanced observers measure is given in [70], which is a hybrid of [31]. It does not require observers to detect DS attacks. Instead, system software performs deep inspection analysis and removes the transaction from the memory pool if double-spending is detected. [53] suggested disabling direct incoming connections of seller that will help seller from not connecting with malicious peers.

Sybil attack can be prevented by increasing the number of confirmed blocks [36], restricting miners to mining

consecutive blocks [79], and applying fees to create node identities [80]. [81] used a mixing technique to reduce the chances of Sybil attack. We have not found any preventive measures against a Balance attack in Bitcoin. Heilman et al. proposed countermeasures against Eclipse attacks, such as accepting random outgoing connections, disabling direct incoming connections, keeping whitelisting of nodes, and using anchor connections [45]. The authors in [82] suggested waiting for more number of confirmed blocks. Ian Miers et al. [66], [83] proposed Zero-coin, a zero-knowledge proof approach that prevents double-spending attacks by reducing verification time. They used commitments and were zero-knowledge-proof, but their proof size was large.

[42] proposed a solution that suggests filtering the transactions by their IP address, using dynamic timeouts, and assigning a penalty to non-responsive nodes to prevent a Tampering attack. To prevent Tampering attacks, a node can announce the time it takes to mine a block together with the advertisement of a new block. This will help another peer in the network to estimate the time required for block mining. As mining time varies across peers, the authors suggested using dynamic timeouts. The routing attack can be prevented by increasing node connectivity by monitoring round trip time as suggested in [48]. The preventive measures against a Vector 76 attack are similar to Race attack measures – disabling direct incoming connections and monitoring outgoing ones [53].

The proposed solutions to safeguard against the Finney attack are to increase confirmed blocks, reduce waiting time to logarithmic values, keep some listening period before giving service, and insert observers into the network [38], [63], [87]. The brute-force attack also has solutions similar to race attacks.

In [88] [89], the authors proposed a method to prevent the formation of large mining pools called two-phase proof-of-work. The Selfish mining attack can be prevented by using the truth state for the block as given in [91]. It is a strategy to analyze selfish behavior based on block height. The difference between succeeding block heights and the average expected block height would gain more truth state; this provides a great advantage to honest miners. The authors suggested using an anti-bribing scheme to prevent bribery attacks, paying pool miners more than what the attacker is paying. The counter-bribing process cannot be active for more extended periods because it makes the PoW puzzle harder for pool miners as time passes [56]. Authors in [60] suggest that pool managers should involve miners who are known to them personally and trusted. The pool manager can close a pool if the pool's earnings go down below the levels expected from its computational effort. [58] suggested using cryptographic commitment schemes to prevent a block withholding attack.

6. Observations and Future Directions

We studied and analyzed recent literature related to double-spending (DS) attacks in the Bitcoin blockchain. It helped us categorize DS attacks based on Bitcoin protocol

TABLE 6: A summary of attacks and preventive countermeasures related to Module 1.

Attacks and preventive countermeasures			
Attacks	Existing countermeasures	Approach Type	References
Race	Increase the minimum number of confirmations	Forwarding	[5], [26]
	Insert observers in the network, Communicate alerts among peers	Conceptual, Broadcasting	[31]
	Disable direct incoming connection	Inform	[53]
	Nearby peers should notify against double-spending as early as possible	Inform, Forwarding	[52]
	Enhanced Observers, neutral observers	Detection, Observer, monitor	[70], [84]
Sybil	increase confirmed blocks	Detection	[36]
	Restrict to mine consecutive blocks	Conceptual	[79]
	Fee to create node identity	Detection	[80]
	2 party mixing protocol Xim	Detection	[81]
	charging an identity fees and setting a deadline	detection	[85]
Balance			[44]
Eclipse	Increase confirmed blocks	Detection	[82]
	Disable incoming connection and random outgoing connection	Inform, Conceptual	[45]
	Deterministic random eviction and whitelist nodes	Detection, Forwarding	[45]
	Feeler connections, anchor connections	Inform	[45]
	detect suspicious block timestamps and gossip blockchain views to contacted server, their clients	Detection, Broadcast	[86]
Tampering	Improve block request management system, randomly choose sender, filter transactions by IP address, dynamic timeouts, penalizing non-responding nodes	Inform, Detection	[42]
Routing	increase the diversity of node connections, monitor RTT, use gateways in different Ases	Monitor, Inform	[48]
Vector76	disable direct incoming connections, monitor outgoing node connections	Inform, Monitor	[53]

modules. Mining and propagation-synchronization modules of the Bitcoin protocol are the most vulnerable. They affect time and cost per transaction and per block, which vary based on the network bandwidth, node connectivity, and the computational power of a node itself. Also, we observed that most of the literature considered one attacker and multiple honest nodes in their network model. However, it is important to explore the area from another viewpoint to understand how multiple malicious nodes (attackers) perform a DS attack.

Few of these DS attacks cause blockchain forks. The traditional PoW mechanism suggested by Nakamoto inherently allows forking, which increases the probability of a successful DS attack. But forking leads to destroying the correctness, integrity, and consistency of the entire ledger. Hence, severe economic losses to users can occur. The current Bitcoin network does not restrict miners who want to launch DS attacks. As every mining node has similar functionalities, it is very difficult to find which mining node is malicious. There is a lack of effective mechanisms that can help us recognize if a miner (user) has created this kind of attack. We observed that it is important to restrict users' capabilities to prevent DS attacks. We suggest an improvement in the Bitcoin protocol by grouping nodes based on their geographical locations. It will help users to connect to peers faster. Also, the protocol needs an im-

provement in the consensus mechanism and fork-resolving policies. We also show combined attacks in Figure 6. This figure represents which attacks can be combined to harm the system more by double-spending the amount from the transactions. In the future, we will analyze the probability of success of a double-spending attack and evaluate these models' performance and the profit of the attackers.

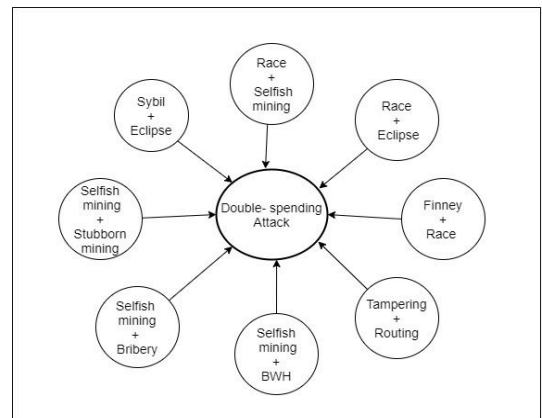


Figure 6: Combined Attacks causing Double-spending

TABLE 7: A summary of attacks and preventive countermeasures related to Module 2.

Attacks and preventive countermeasures			
Attacks	Existing countermeasures	Approach Type	References
Finney	Increase confirmed blocks	Detection	[38]
	Logarithmic waiting times	Conceptual	[63]
	Listening period and insert observers	Conceptual	[87]
Bruteforce	inserting observers	Monitor, Broadcast	[31]
	notify the seller about an on-going double spend	Inform	[53]
Goldfinger or >50%	Disincentivize large mining pools	Broadcast	[88], [89]
	block access restriction	Detection	[90]
Selfish Mining	truth state for blocks at the fork instant to identify selfish mining behaviors	Detection, Monitor	[91]
	PDA mechanism decreasing the probability of consecutive winning	Conceptual	[92]
	Broadcasting intermediate blocks and add transparency to the mining process, block acceptance flag	Broadcast	[93] [94]
	Weighted fork resolving policy and reward distribution policy	Conceptual	[95]
Bribery	increase the rewards for honest miners, make aware the miners to the long-term losses of bribery	Detection	[56]
Block Withholding	include only known and trusted miners in pool, dissolve and close a pool when revenue drops from expected	Conceptual	[60]
	Cryptographic commitment schemes	Conceptual	[58]
	infiltration detection and reduce the profit of attackers pool to decrease the damage	Detection	[96]

7. Conclusion

In this paper, we focused on the double-spending problem in the Bitcoin protocol. By studying the theoretical and experimental work, we systematically examined the possible ways to double-spend bitcoins and their association with delay-causing factors present in the Bitcoin protocol. We analyzed all double-spending vulnerabilities and their causes and possible consequences. We discussed the risks associated with each factor of the protocol and preventive measures against double-spending attacks. Finally, we presented the summary of Bitcoin security enhancement.

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